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RECURRENCE RELATIONS

A formula which defines any term of a sequence in terms of any no. of its previous terms or which expresses any term of a sequence as a function of its previous terms is called recursive and the relation is called a recurrence relation.

For eg, the n^{th} terms a_n of the sequence 3, 8, 13, 18, 23, can be expressed as,

$$a_n = a_{n-1} + 5, \text{ for } n \geq 2 \text{ and } a_1 = 3.$$

This formula is called recursive formula and the relation is called the recurrence relation (some times called difference equation)

Order of the Recurrence Relation

Order of $a_n = F(a_{n-1}, a_{n-2}, \dots, a_{n-k}, n)$ is k , where a_n depends on one or more of the previous k terms and k is the smallest such integer.

We do not define an order for recurrence relations of the form $a_n = F(a_{n-1}, a_{n-2}, \dots, n)$ that depends on each of its previous terms.

Degree of a Recurrence Relation

For eg: $a_n = 3a_{n-1} + n^2$ is non homogeneous of order 1 & degree 1.

$a_n = n a_{n-2} + 2^n$ is non homogeneous of order 2 & degree 1.

$a_n = a_{n-1} + a_{n-2} + \dots + a_0$ is homogeneous of no order & degree 1.

Methods of Solving Recurrence Relations

1. Substitution or Iteration method.
2. Characteristic roots method.
3. Generating functions method.

1. Substitution Method (or Iteration method)

In this method we start with the first term and terminate at the n^{th} term.

Ex 1. Find an explicit formula for the sequence defined by a recurrence relation $a_n = a_{n-1} + 3$ with $a_1 = 2$.

Sol Putting $n = 2, 3, \dots, n$ in the given recurrence relation

$$a_2 = a_1 + 3$$

$$a_3 = a_2 + 3 = (a_1 + 3) + 3 = a_1 + 2 \cdot 3$$

$$a_4 = a_3 + 3 = (a_1 + 2 \cdot 3) + 3 = a_1 + 3 \cdot 3$$

$\vdots$

$$a_n = a_{n-1} + 3 = (a_1 + (n-2) \cdot 3) + 3 = a_1 + 3n - 3$$

$$= a_1 + (n-1) \cdot 3$$

$$= 2 + (n-1) \cdot 3 \quad (\text{on putting } a_1 = 2)$$

$$a_n = 3n - 1 \quad \underline{\text{Ans.}}$$

Ex 2. Find an explicit formula for the sequence defined by the recurrence relation $b_n = 2b_{n-1} + 1$ with I.C. $b_1 = 7$.

Sol. Putting $n = 2, 3, \dots$ in the given recurrence relation

$$b_2 = 2b_1 + 1$$

$$b_3 = 2b_2 + 1 = 2(2b_1 + 1) + 1 = 2^2 b_1 + 2^1 + 2^0$$

$$b_4 = 2b_3 + 1 = 2(2^2 b_1 + 2^1 + 2^0) + 1$$

$$= 2^3 b_1 + 2^2 + 2^1 + 2^0$$

$\vdots$

$$b_n = 2b_{n-1} + 1 = 2^{n-1} b_1 + 2^{n-2} + 2^{n-3} + \dots + 2^1 + 2^0$$

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$$= 2^{n-1} \cdot b_1 + \frac{(2^{n-1} - 1)}{2-1}$$

(on summing G.P.)

$$= 2^{n-1} b_1 + 2^{n-1} - 1$$

$$= 7 \cdot 2^{n-1} + 2^{n-1} - 1. \quad \underline{\text{Ans.}}$$

Ex. 3 Solve the recurrence relation

$$a_n = a_{n-1} + 2, \quad n \geq 2 \text{ with } a_1 = 3.$$

Iteration Method

Ex.

$$a_k = 2a_{k-1} \quad a_0 = 1$$

$$a_k = 2a_{k-1}$$

$$= 2 \cdot 2a_{k-2} = 2^2 a_{k-2}$$

$$= 2 \cdot 2^2 a_{k-3} = 2^3 a_{k-3}$$

$$\dots 2^k a_{k-k} = 2^k a_0 = 2^k \cdot 1 = 2^k$$

$$a_k = 2^k \quad \forall k \geq 1$$

$$a_1 = 4$$

Ex.

$$a_n = a_{n-1} + n$$

$$a_n = a_{n-1} + n$$

$$= a_{n-2} + (n-1) + n$$

$$= a_{n-2} + n + (n-1)$$

$$= a_{n-3} + n + (n-1) + (n-2)$$

$$a_n = a_{n-k} + (n) + (n-1) + (n-2) + \dots + (n-(k-1))$$

for $k = n-1$

$$a_n = a_{n-(n-1)} + n + (n-1) + (n-2) + \dots + (n-((n-1)-1))$$

$$= a_1 + n + (n-1) + (n-2) + \dots + 2$$

$$(= a_1 + n + (n-1) + (n-2) + \dots + 2 + 1 - 1$$

$$= a_1 + \frac{n(n-1)}{2} - 1$$

$$a_1 = 4$$

$$= 3 + \frac{n(n-1)}{2}$$

Ex.

$$a_n = n a_{n-1} \quad a_0 = 5$$

$$a_n = n a_{n-1}$$

$$= n(n-1) a_{n-2}$$

$$= n(n-1)(n-2) a_{n-3}$$

.....

$$= n(n-1)(n-2) \dots (n-k) a_{n-(k+1)}$$

for $k = n-1$

$$a_n = n(n-1)(n-2) \dots 1 a_{n-(n-1+1)}$$

$$= n! a_0 = 5n!$$

Ex.

$$a_n = 2 a_{n-1} + 1 \quad a_1 = 7$$

$$a_n = 2 a_{n-1} + 1$$

$$= 2(2 a_{n-2} + 1) + 1 = 2^2 a_{n-2} + 2 + 1$$

$$= 2^2(2 a_{n-3} + 1) + 2 + 1$$

$$= 2^3 a_{n-3} + 2^2 + 2 + 1$$

.....

$$a_n = 2^k a_{n-k} + 1 + 2 + 2^2 + \dots + 2^{k-1}$$

for $k = n-1$

$$a_n = 2^{n-1} a_1 + 1 + 2 + 2^2 + \dots + 2^{n-2}$$

$$= 2^{n-1} a_1 + \frac{1(2^{n-1} - 1)}{2 - 1}$$

$$= 7 \cdot 2^{n-1} + 2^{n-1} - 1 = 2^{n+2} - 1$$

The Method of Characteristic Roots

In this method the solution is obtained as the sum of two parts, the homogeneous solution, which satisfy the recurrence relation when the right hand of the relation is set to 0 i.e., $f(n) = 0$, and the particular solution, which satisfies the relation with $f(n)$ on the right hand side.

(a) Homogeneous Recurrence Relation

The basic approach for solving homogeneous relation [$f(n) = 0$] is to look for solutions of the form $a_n = r^n$. Now $a_n = r^n$ is a solutions of the recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} \quad \text{if \& only if}$$

$$r^n = c_1 r^{n-1} + c_2 r^{n-2} + \dots + c_k r^{n-k}$$

Dividing both sides by r^{n-k}

$$r^k = c_1 r^{k-1} + c_2 r^{k-2} + \dots + c_k$$

$$\text{or} \quad r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_k = 0$$

which is called the characteristic or auxiliary eqⁿ of the recurrence relation. The solⁿ of this eqⁿ are called the characteristic roots of the recurrence relation. A characteristic eqⁿ of the kth degree has k characteristic roots.

Distinct roots : If the characteristic eqⁿ has distinct roots $r_1, r_2, \dots, r_k$, then the general form of the solⁿs for homogeneous eqⁿ is

$$a_n = b_1 r_1^n + b_2 r_2^n + b_3 r_3^n + \dots + b_k r_k^n \quad \text{where } b_1, b_2, \dots, b_k$$

are constants which may be chosen to satisfy any initial conditions.

Ex.

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Ex.1. Solve $a_n = a_{n-1} + 2a_{n-2}$, $n \geq 2$ with the I.C. $a_0=0, a_1=1$.

Sol $a_n = a_{n-1} + 2a_{n-2}$

i.e. $a_n - a_{n-1} - 2a_{n-2} = 0$ ----- (1)

is a second order linear homogeneous recurrence relation with constant coefficients

Let $a_n = r^n$ is a solⁿ of (1). The characteristic eqⁿ is

$$r^2 - r - 2 = 0$$

$$(r-2)(r+1) = 0$$

or $r = 2, -1$ distinct real roots, So the general solⁿ is

$$a_n = b_1 2^n + b_2 (-1)^n$$

Again $a_0 = 0 \Rightarrow b_1 + b_2 = 0$

And $a_1 = 1 \Rightarrow 2b_1 - b_2 = 1$

After solving these eqⁿs

$$b_1 = \frac{1}{3}, b_2 = -\frac{1}{3}$$

Hence the explicit solⁿ is

$$a_n = \frac{1}{3}(2)^n - \frac{1}{3}(-1)^n \quad \underline{\underline{\text{Ans}}}$$

Multiple Roots: If the characteristic eqⁿ of a homogeneous recurrence relation is $(r-2)^3 = 0$, then $r=2$ is a required root of multiplicity 3. Then the general solⁿ is

$$a_n = (b_1 + nb_2 + n^2b_3) 2^n$$

In general, if r is a root of the characteristic eqⁿ of m th order of a given recurrence relation with multiplicity m , then the general form of the solⁿ is

$$a_n = (b_1 + nb_2 + n^2b_3 + \dots + n^{m-1}b_m) r^n$$

where $b_1, b_2, \dots, b_m$ are constants which may be chosen to satisfy any initial conditions.

Mixed Roots: A combination of distinct and multiple roots is also possible i.e., some roots of a characteristic eqⁿ are distinct and some roots are equal. If the characteristic eqⁿ of a homogeneous recurrence relation of 5th order is $(r-2)(r-4)(r-3)^3=0$, then $r=2, 4, 3, 3, 3$. Then the general solⁿ is

$$a_n = b_1 2^n + b_2 4^n + (b_3 + nb_4 + n^2 b_5) 3^n$$

Ex.2. Solve the recurrence relation

$$a_n = 4(a_{n-1} - a_{n-2}) \text{ with initial condⁿ } a_0 = a_1 = 1$$

Sol $a_n - 4a_{n-1} + 4a_{n-2} = 0$ ----- (1)

Let $a_n = r^n$ is a solⁿ of (1).

Then the characteristic eqⁿ is $r^2 - 4r + 4 = 0$

$$\Rightarrow (r-2)^2 = 0 \Rightarrow r = 2, 2.$$

Thus the general solⁿ is

$$a_n = (b_1 + nb_2) 2^n$$

So, $a_0 = 1$, gives $b_1 = 1$

and $a_1 = 1$ gives $2(b_1 + b_2) = 1 \Rightarrow b_2 = -1/2$.

So, the required solⁿ is

$$a_n = \left(1 - \frac{1}{2}n\right) 2^n. \text{ Ans.}$$

Ex.3. Solve the recurrence relation

$$a_n - 8a_{n-1} + 21a_{n-2} - 18a_{n-3} = 0$$

Sol Let $a_n = r^n$ be a solⁿ of the given eqⁿ.

The characteristic eqⁿ is

$$r^3 - 8r^2 + 21r - 18 = 0$$

$$r^2(r-2) - 6r(r-2) + 9(r-2) = 0$$

$$(r-2)(r^2 - 6r + 9) = 0$$

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$$(r-2)(r-3)^2 = 0$$

$$\Rightarrow r = 2, 3, 3.$$

So the general solⁿ is

$$a_n = (b_1 + nb_2)3^n + b_3 2^n. \underline{\text{Ans}}$$

(b) Non Homogeneous Recurrence Relation

A second order non homogeneous linear recurrence relation with constant coefficients is of the form

$$a_{n-2} + 5a_{n-1} + 6a_n = f(n)$$

Its solⁿ a_n consists of two parts

1. Homogeneous solⁿ $a_n^{(h)}$ of the given recurrence relation by keeping $f(n) = 0$.
2. Particular solⁿ $a_n^{(p)}$ of the given recurrence relation with $f(n)$ on the right hand side.

The required general solⁿ is

$$a_n = a_n^{(h)} + a_n^{(p)}$$

Particular Solution

Form of $f(n)$

b^n (if b is not a root of characteristic eqⁿ) Polynomial $P(n)$ of degree m

$c^n P(n)$ (if c is not a root of ch. eqⁿ)

b^n (if b is a root of characteristic eqⁿ of multiplicity s)

$c^n P(n)$ (if c is a root of characteristic eqⁿ of multiplicity t)

$\sin bn$ or $\cos bn$

$b^n \sin bn$ or $b^n \cos bn$

Trial Function

$A b^n$

$$A_0 + A_1 n + A_2 n^2 + \dots + A_m n^m$$

$$c^n (A_0 + A_1 n + \dots + A_m n^m)$$

$$A_0 n^s b^n$$

$$n^t (A_0 + A_1 n + A_2 n^2 + \dots + A_m n^m) c^n$$

$$A_0 \sin bn + A_1 \cos bn$$

$$b^n (A_0 \sin bn + A_1 \cos bn)$$

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NOTE: (i) If $f(n)$ is constant i.e., a polynomial of degree zero, the trial sol^n is taken as A .

(ii) If $f(n)$ is a linear combination of the above forms, the trial sol^n is taken as the sum of corresponding trial functions with different unknown constant coefficients to be determined.

Type 1: When $f(n) = \alpha$, a constant

Ex 1. Solve $a_{n+2} - 5a_{n+1} + 6a_n = 2$ with I.C. $a_0 = 1$ & $a_1 = -1$.

Sol The associated homogeneous recurrence relation is

$$a_{n+2} - 5a_{n+1} + 6a_n = 0 \quad \text{--- (1)}$$

Let $a_n = r^n$ be a sol^n of (1).

$$\begin{aligned} \text{The characteristic eq}^n \text{ is } r^2 - 5r + 6 &= 0 \\ \Rightarrow (r-3)(r-2) &= 0 \\ \Rightarrow r &= 3, 2. \end{aligned}$$

So, the sol^n of (1) is $a_n^{(h)} = C_1 3^n + C_2 2^n$.

To find the particular sol^n of the given eq^n , Let $a_n = A$.
Substituting in the given eq^n .

$$A - 5A + 6A = 2 \Rightarrow A = 1.$$

$\therefore$ Particular sol^n $a_n^{(p)} = 1$

Hence the general sol^n is $a_n = a_n^{(h)} + a_n^{(p)}$

$$= C_1 3^n + C_2 2^n + 1 \quad \text{--- (2)}$$

To find C_1 & C_2 , put $n=0$ & $n=1$ in (2)

$$a_0 = C_1 + C_2 + 1$$

$$C_1 + C_2 = 0 \quad \text{--- (3)}$$

Again,

$$a_1 = 3C_1 + 2C_2 + 1$$

$$3C_1 + 2C_2 = -2 \quad \text{--- (4)}$$

Solving (3) & (4), $C_1 = -2$ and $C_2 = 2$

From (2),

$$a_n = -2 \cdot 3^n + 2 \cdot 2^n + 1$$

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(12) When $f(n) = X$ is a polynomialEx. 2. Solve the following $y_{n+2} - y_{n+1} - 2y_n = n^2$ --- (1)Sol. Let $y_n = r^n$ be a solⁿ of (1)The characteristic eqⁿ is $r^2 - r - 2 = 0$

OR

$$(r+1)(r-2) = 0 \Rightarrow r = -1, 2$$

The solⁿ of (1) is $y_n^{(h)} = C_1(-1)^n + C_2 \cdot 2^n$ Let the particular solⁿ of the given eqⁿ be

$$y_n = A_0 + A_1 n + A_2 n^2 \quad (\text{since } f(n) \text{ is a polynomial of degree 2})$$

Substituting in the given eqⁿ,

$$A_0 + A_1(n+2) + A_2(n+2)^2 - [A_0 + A_1(n+1) + A_2(n+1)^2] - 2[A_0 + A_1 n + A_2 n^2] = n^2$$

$$\text{or, } (-2A_0 + A_1 + 3A_2) + (-2A_1 + 2A_2)n - 2A_2 n^2 = n^2$$

On comparing the coefficients of like powers of n ,

$$-2A_0 + A_1 + 3A_2 = 0 \quad \text{--- (1)}$$

$$-2A_1 + 2A_2 = 0 \quad \text{--- (2)}$$

$$-2A_2 = 1 \quad \text{--- (3)}$$

$$\text{From (3), } A_2 = -1/2$$

$$\text{From (2), } A_1 = A_2 = -1/2$$

$$\text{From (1), } -2A_0 - \frac{1}{2} - \frac{3}{2} = 0 \Rightarrow A_0 = -1$$

Therefore, particular solⁿ is

$$y_n^{(p)} = -1 - \frac{1}{2}n - \frac{1}{2}n^2$$

Hence the general solⁿ is

$$y_n = y_n^{(h)} + y_n^{(p)}$$

$$y_n = C_1(-1)^n + C_2 \cdot 2^n - 1 - \frac{n}{2} - \frac{n^2}{2}$$

Type 2: When $f(n)$ is of the form an^d , where a is constant and $d \in \mathbb{N}$

Rule 2: A particular solⁿ of a recurrence relation with non-homogeneous part, $f(n) = an^d$, where a is a known constant and $d \in \mathbb{N}$, is of the form:

(i) $A_0 + A_1 n + A_2 n^2 + \dots + A_d n^d$, if 1 is not a characteristic root.

(ii) $A_0 n^m + A_1 n^{m+1} + A_2 n^{m+2} + \dots + A_d n^{m+d}$, if 1 is a characteristic root of r.r. with multiplicity m , where $A_0, A_1, A_2, \dots, A_d$ are constants.

Ex 3. Find complete solution of the recurrence relation

$$u_n = 3u_{n-1} - 4n \quad (n \geq 1)$$

Sol. The associated homogeneous recurrence relation is

$$u_n - 3u_{n-1} = 0 \quad \text{--- (1)}$$

The characteristic eqⁿ is $r - 3 = 0$

$$\Rightarrow r = 3$$

Therefore the general solⁿ of (1) is

$$u_n = b_1 \cdot 3^n$$

Now find the particular solⁿ.

Here $f(n) = -4n = -4 \cdot n^1$ where $d=1$. Also 1 is not a characteristic root of (1), the particular solⁿ is of the form

$$u_n = A_0 + A_1 n$$

Putting this value in the given recurrence relation

$$A_0 + A_1 n = 3[A_0 + A_1(n-1)] - 4n$$

$$-2A_0 + 3A_1 - 2A_1 n = -4n$$

Comparing coefficients of n in both sides

$$-2A_0 + 3A_1 = 0 \quad \& \quad -2A_1 = -4$$

$$\boxed{A_0 = 3}$$

$$\Rightarrow \boxed{A_1 = 2}$$

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Thus the particular solⁿ becomes

$$u_n^{(b)} = 3 + 2n$$

∴ The complete solⁿ is $u_n = u_n^{(h)} + u_n^{(b)}$

$$u_n = b_1 \cdot 3^n + 3 + 2n \quad \underline{\underline{\text{Ans.}}}$$

Ex.3. Find the complete solⁿ of $u_n = u_{n-1} + n$, $n \geq 0$, $u_1 = 3$.

Sol. The given recurrence relation is

$$u_n = u_{n-1} + n, \quad n \geq 0, \quad u_1 = 3.$$

The associated homogeneous recurrence relation is

$$u_n - u_{n-1} = 0$$

The characteristic eqⁿ is $r - 1 = 0 \Rightarrow r = 1$

The general solⁿ is $u_n^{(h)} = b_1 \cdot 1^n = b_1$

To find particular solⁿ, we have $f(n) = n$ where $d=1$
As the characteristic root is 1, the particular solⁿ form

is $u_n = A_0 n + A_1 n^2$

Putting value of u_n in given recurrence relation

$$A_0 n + A_1 n^2 = A_0 (n-1) + A_1 (n-1)^2 + n$$

$$A_0 n + A_1 n^2 = (-A_0 + A_1) + (A_0 - 2A_1)n + A_1 n^2 + n$$

OR. $(A_0 - A_1) + 2A_1 n = n$

Comparing the coefficients of n^0 and n^1 in both sides

$$A_0 - A_1 = 0 \quad \text{and} \quad 2A_1 = 1$$

$$\boxed{A_0 = \frac{1}{2}}$$

$$\boxed{A_1 = \frac{1}{2}}$$

Thus the particular solⁿ is

$$u_n^{(b)} = \frac{1}{2}n + \frac{1}{2}n^2$$

∴ The complete solⁿ is

$$u_n = b_1 + \frac{1}{2}n + \frac{1}{2}n^2$$

The initial condition is $u_1 = 3$,

we get $3 = b_1 + 1$ or $b_1 = 2$.

The required solⁿ is

$$u_n = 2 + \frac{1}{2}n + \frac{1}{2}n^2, \quad n \geq 1. \quad \text{Ans.}$$

Type 3: When $f(n)$ is of the form ar^n , where a is a known constant.

Rule 3: (i) Ar^n , if r is not a characteristic root.
(ii) $A \cdot n^m r^n$, if r is a characteristic root with multiplicity m ,

Ex. 4. Find complete solⁿ of the recurrence relation

$$u_n - 4u_{n-1} + 3u_{n-2} = 5^n$$

Sol. The associated homogeneous eqⁿ is

$$u_n - 4u_{n-1} + 3u_{n-2} = 0 \quad \text{--- (1)}$$

The characteristic eqⁿ is

$$r^2 - 4r + 3 = 0 \quad \text{or} \quad (r-3)(r-1) = 0$$

Its characteristic roots are

$$r = 3, 1$$

Therefore the general solⁿ of (1) is

$$u_n^{(h)} = b_1 \cdot 3^n + b_2 (1)^n = b_1 \cdot 3^n + b_2$$

To find particular solⁿ we see that $f(n) = 5^n$, $a = 1$, $r = 5$.

Moreover $r = 5$ is not a characteristic root, therefore the particular solⁿ

$$u_n = A \cdot 5^n \quad \text{--- (2)}$$

Putting this value of u_n in given recurrence relation,

$$A \cdot 5^n - 4 \cdot A \cdot 5^{n-1} + 3 \cdot A \cdot 5^{n-2} = 5^n$$

or dividing by 5^{n-2} ,

$$A \cdot 5^2 - 4 \cdot A \cdot 5 + 3 \cdot A = 25$$

$$\Rightarrow A = 25/8.$$

Putting this value of A in (2) the particular solⁿ becomes

$$u_n^{(p)} = \frac{25}{8} \cdot 5^n = \frac{5^{n+2}}{8}$$

The complete solⁿ is

$$u_n = b_1 \cdot 3^n + b_2 + \frac{5^{n+2}}{8}. \quad \text{Ans.}$$

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Ex. Solve the recurrence relation

$$u_n = 6u_{n-1} - 9u_{n-2} + 3^n$$

Sol. The associated homogeneous eqⁿ is

$$u_n - 6u_{n-1} + 9u_{n-2} = 0 \quad \text{--- (1)}$$

The characteristic eqⁿ is $r^2 - 6r + 9 = 0$

$$\Rightarrow (r-3)^2 = 0$$

The characteristic root is $r=3$ of multiplicity 2.Therefore the general solⁿ of (1) is

$$u_n = (b_1 + b_2 n) 3^n$$

To find the particular solⁿ we see that $f(n) = 3^n$ i.e., ar^n , $a=1$, $r=3$ and 3 is also characteristic root.Therefore the particular solⁿ is

$$u_n = A \cdot n^2 \cdot 3^n \quad \text{--- (2)}$$

Putting this value of u_n in eqⁿ (1)

$$A \cdot n^2 \cdot 3^n = 6A \cdot (n-1)^2 \cdot 3^{n-1} - 9A \cdot (n-2)^2 \cdot 3^{n-2} + 3^n$$

Dividing both sides by 3^{n-2}

$$A \cdot n^2 \cdot 3^2 = 6A \cdot (n-1)^2 \cdot 3 - 9A \cdot (n-2)^2 + 3^2$$

$$9A \cdot n^2 = 18A \cdot (n-1)^2 - 9A \cdot (n-2)^2 + 9$$

$$A [9n^2 - 18(n-1)^2 + 9(n-2)^2] = 9$$

$$18A = 9 \Rightarrow A = \frac{1}{2}$$

Putting this value of A in eqⁿ (2), the particular solⁿ is

$$u_n^{(p)} = \frac{n^2 \cdot 3^n}{2}$$

The required solⁿ is $u_n = (b_1 + b_2 n) 3^n + \frac{n^2 \cdot 3^n}{2} \quad \underline{\underline{\text{Ans}}}$ Type 4: When $f(n)$ is of the form $f_1(n) + f_2(n)$.If $\{a_n\}$ is a solⁿ of $u_n = c_1 u_{n-1} + c_2 u_{n-2} + \dots + c_k u_{n-k} + f_1(n)$ and $\{b_n\}$ is a solⁿ of $u_n = c_1 u_{n-1} + c_2 u_{n-2} + \dots + c_k u_{n-k} + f_2(n)$ then the solⁿ of $u_n = c_1 u_{n-1} + \dots + c_k u_{n-k} + A f_1(n) + B f_2(n)$ is given by $A \cdot a_n + B \cdot b_n$ where A & B are constants.

Ex. Solve the recurrence relation

$$u_n - 7u_{n-1} + 16u_{n-2} - 12u_{n-3} = 2^n + 3^n, \quad u_0 = 0, u_1 = 1, u_2 = 0$$

sol. Associated homogeneous relation shall be

$$u_n - 7u_{n-1} + 16u_{n-2} - 12u_{n-3} = 0 \quad \dots (1)$$

The characteristic eqⁿ is

$$r^3 - 7r^2 + 16r - 12 = 0 \quad \text{or} \quad (r-2)^2 (r-3) = 0$$

The characteristic roots are

$$r = 2 \text{ of multiplicity } m = 2$$

$$\text{and } r = 3 \quad " \quad m = 1$$

The general solⁿ of (1) is

$$u_n^{(h)} = (b_1 + b_2 n) 2^n + b_3 \cdot 3^n$$

The non-homogeneous part $f(n)$ is of the form

$$f(n) = f_1(n) + f_2(n) = 2^n + 3^n$$

To find the particular solution corresponding to $f_1(n) = 2^n$ i.e. $a r^n$ where $a = 1$, $r = 2$ and also 2 is a characteristic root of multiplicity $m = 2$.

Therefore, the particular solⁿ corresponding to $f_1(n) = 2^n$ is of the form $A_1 n^m r^n = A_1 n^2 \cdot 2^n$

Again to find particular solⁿ corresponding to $f_2(n) = 3^n$ i.e. $a r^n$ where $a = 1$, $r = 3$ and also 3 is a characteristic root of multiplicity $m = 1$.

Therefore particular solⁿ is of the form $A_2 n^m r^n = A_2 n \cdot 3^n$

Therefore the particular solⁿ is

$$u_n = A_1 n^2 2^n + A_2 n 3^n \quad \dots (2)$$

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Putting this value of u_n in given r.r.

$$A_1 n^2 \cdot 2^n + A_2 \cdot n \cdot 3^n - 7 [A_1 (n-1)^2 \cdot 2^{n-1} + A_2 (n-1) \cdot 3^{n-1}] \\ + 16 [A_1 (n-2)^2 \cdot 2^{n-2} + A_2 (n-2) \cdot 3^{n-2}] - 12 [A_1 (n-3)^2 \cdot 2^{n-3} \\ + A_2 (n-3) \cdot 3^{n-3}] = 2^n + 3^n$$

$$\text{or } A_1 [n^2 \cdot 2^n - 7(n-1)^2 \cdot 2^{n-1} + 16(n-2)^2 \cdot 2^{n-2} - 12(n-3)^2 \cdot 2^{n-3}] \\ + A_2 [n \cdot 3^n - 7(n-1) \cdot 3^{n-1} + 16(n-2) \cdot 3^{n-2} - 12(n-3) \cdot 3^{n-3}] = 2^n + 3^n$$

$$\text{or } A_1 \cdot 2^{n-3} [n^2 \cdot 2^3 - 7(n-1)^2 \cdot 2^2 + 16(n-2)^2 \cdot 2 - 12(n-3)^2] \\ + A_2 \cdot 3^{n-3} [n \cdot 3^3 - 7(n-1) \cdot 3^2 + 16(n-2) \cdot 3 - 12(n-3)] = 2^n + 3^n$$

$$\text{or } A_1 \cdot 2^{n-3} [-8] + A_2 \cdot 3^{n-3} [3] = 2^n + 3^n$$

$$\text{or } -A_1 \cdot 2^n + A_2 \cdot 3^{n-2} = 2^n + 3^n$$

Comparing the coefficients,

$$\boxed{A_1 = -1}, \quad \boxed{A_2 = 9}$$

Putting these values in eq (2), the particular solⁿ is

$$u_n^{(p)} = -1 \cdot n^2 \cdot 2^n + 9n \cdot 3^n$$

Hence the complete solⁿ is

$$u_n = (B_1 + B_2 n) 2^n + B_3 \cdot 3^n - n^2 \cdot 2^n + 9n \cdot 3^n$$

$$\text{or } u_n = (B_1 + B_2 n - n^2) 2^n + (B_3 + 9n) 3^n$$

The I.C. are $u_0 = 0, u_1 = 1, u_2 = 0$

$$B_1 + B_3 = 0$$

$$1 = (B_1 + B_2 - 1) \cdot 2 + (B_3 + 9) \cdot 3 \quad \text{or } 2B_1 + 2B_2 + 3B_3 = -24$$

$$0 = (B_1 + 2B_2 - 4) \cdot 4 + (B_3 + 18) \cdot 9 \quad \text{or } 4B_1 + 8B_2 + 9B_3 = -146$$

Solving these three eq^s,

$$B_1 = 50, \quad B_2 = 13, \quad B_3 = -50$$

Therefore the required complete solⁿ is

$$u_n = (50 + 13n - n^2) 2^n + (9n - 50) 3^n \quad \underline{\text{Ans.}}$$

Characteristic Roots

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As sum of 2 parts.

Homogenous Sol.

Particular Sol

$$RHS = 0$$

$$RHS = f(n)$$

$$a_n + C_1 a_{n-1} + C_2 a_{n-2} + \dots + C_k a_{n-k} = 0$$

Homo RR

put $a_n = r^n$

$$r^n + C_1 r^{n-1} + C_2 r^{n-2} + \dots + C_k r^{n-k}$$

Divide by r^{n-k}

$$r^k + C_1 r^{k-1} + C_2 r^{k-2} + \dots + C_k = 0$$

Characteristic Eqⁿ.

& its roots are char roots.

Let char roots are $r_1, r_2, r_3, \dots, r_k$

Case 1 Real & Distinct Roots.

gen sol $a_n = b_1 r_1^n + b_2 r_2^n + \dots + b_k r_k^n$

$b_1, b_2, \dots$ Const which are obtained by using init conditions.

Case 2 Multiple roots $r_1 = r_2 = r_3, r_4, \dots, r_k$

then gen sol

$$a_n = (b_1 + b_2 n + b_3 n^2) r_1^n + b_4 r_4^n + \dots$$

Case 3 Imaginary roots $r_1 \pm i r_2$

gen sol

$$a_n = b_1 (r_1 + i r_2)^n + b_2 (r_1 - i r_2)^n$$

Case, imp. repeated roots

$$q_r = (b_1 + b_2 r) (r_1 + i r_2)^r + (b_3 + b_4 r) (r_1 - i r_2)^r$$

Ex. $q_r - 3q_{r-1} + 2q_{r-2} = 0$

Let $q_r = \alpha^r$

$$\alpha^r - 3\alpha^{r-1} + 2\alpha^{r-2} = 0$$

Divide by α^{r-2}

$$\alpha^2 - 3\alpha + 2 = 0 \quad \text{Char. eqn.}$$

$$(\alpha - 2)(\alpha - 1) = 0 \quad \alpha = 1, 2$$

So $q_r = b_1 + b_2 2^r$

Ex. $q_r - 4q_{r-1} + 4q_{r-2} = 0 \quad r \geq 2 \quad q_0 = \frac{5}{2}, q_1 = 8$

$q_r = \alpha^r$

$$\alpha^r - 4\alpha^{r-1} + 4\alpha^{r-2} = 0$$

Div by α^{r-2}

$$\alpha^2 - 4\alpha + 4 = 0 \quad (\alpha - 2)^2 = 0 \quad \alpha = 2, 2$$

So homo sol.

$$q_r = (b_1 + b_2 r) 2^r$$

$$q_0 = \frac{5}{2}$$

$$\frac{5}{2} = b_1$$

$$q_1 = 8$$

$$8 = (b_1 + b_2) 2$$

$$b_2 = \frac{3}{2}$$

So homo. sol $\Rightarrow q_r = \left(\frac{5}{2} + \frac{3}{2} r \right) 2^r$

Ex. $a_r - 4a_{r-1} + 4a_{r-2} = 0 \quad r \geq 2 \quad a_0 = 5/2 \quad a_1 = 8.$

$$a_r = x^r \quad x^r - 4x^{r-1} + 4x^{r-2} = 0$$

Div by ~~a~~ x^{r-2} .

$$x^2 - 4x + 4 = 0 \quad (x-2)^2 = 0$$

roots are $2, 2$

$$a_r = (b_1 + b_2 r) 2^r$$

$$a_0 = 5/2 \quad r=0 \quad a_0 = 5/2 = (b_1 + b_2 \cdot 0) 2^0 \quad b_1 = 5/2$$

$$a_1 = 8 \quad r=1 \quad a_1 = 8 = (b_1 + b_2 \cdot 1) 2^1$$

$$\left(\frac{5}{2} + b_2\right) 2 = 8$$

$$\frac{5}{2} + b_2 = 4 \quad b_2 = 4 - \frac{5}{2} = \frac{3}{2}$$

So $a_r = \left(\frac{5}{2} + \frac{3}{2}r\right) 2^r$

Ex. $f_n = f_{n-1} + f_{n-2} \quad n \geq 2 \quad f_0 = 1 \quad f_1 = 1.$

$$f_n - f_{n-1} - f_{n-2} = 0$$

$$CE \quad x^2 - x - 1 = 0$$

$$x = \frac{1 \pm \sqrt{5}}{2}$$

$$f_n = b_1 \left(\frac{1+\sqrt{5}}{2}\right)^n + b_2 \left(\frac{1-\sqrt{5}}{2}\right)^n$$

$$f_0 = 1 = b_1 + b_2$$

$$f_1 = 1 = b_1 \left(\frac{1+\sqrt{5}}{2}\right) + b_2 \left(\frac{1-\sqrt{5}}{2}\right)$$

$$= \frac{b_1 + b_2}{2} + \frac{\sqrt{5}}{2} (b_1 - b_2) = \frac{1}{2} + \frac{\sqrt{5}}{2} (b_1 - b_2)$$

$$\frac{1}{2} = \frac{\sqrt{5}}{2} (b_1 - b_2) \quad b_1 - b_2 = \frac{1}{\sqrt{5}} \quad b_1 + b_2 = 1$$

$$b_1 = \frac{1}{2} \left(1 + \frac{1}{\sqrt{5}}\right) = \frac{1+\sqrt{5}}{2\sqrt{5}} \quad b_2 = \frac{1}{2} \left(1 - \frac{1}{\sqrt{5}}\right) = -\left(\frac{1-\sqrt{5}}{2\sqrt{5}}\right)$$

So $f_n = \left(\frac{1+\sqrt{5}}{2\sqrt{5}}\right) \left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2\sqrt{5}}\right) \left(\frac{1-\sqrt{5}}{2}\right)^n$

$$= \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2}\right)^{n+1} - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2}\right)^{n+1}$$

Ex. $a_n - 7a_{n-1} + 16a_{n-2} - 12a_{n-3} = 0$ $a_0 = 1$ $a_1 = 4$ $a_2 = 8$.

CE $x^3 - 7x^2 + 16x - 12 = 0$

Solway $(x-3)(x-2)^2 = 0$ $x = 3, 2, 2$

So:

$$a_n = (b_1 + b_2 n) 2^n + b_3 \cdot 3^n$$

using init cond.

$$1 = b_1 + b_3$$

$$4 = (b_1 + b_2) 2 + 3b_3$$

$$8 = (b_1 + 2b_2) 4 + 9b_3$$

$$\Rightarrow \begin{matrix} b_1 = 5 & b_2 = 3 \\ b_3 = -4 \end{matrix}$$

$$a_n = (5 + 3n) 2^n - 4 (3^n)$$

Ex. $t_{k+4} + 2t_{k+3} + 3t_{k+2} + 2t_{k+1} + t_k = 0$

CE $x^4 + 2x^3 + 3x^2 + 2x + 1 = 0$

$$= (x^2 + x + 1)^2 = 0 \quad x = \frac{-1 \pm \sqrt{3}i}{2} \text{ twice}$$

$$t_k = (b_1 + b_2 k) \left(\frac{-1 + \sqrt{3}i}{2} \right)^k + (b_3 + b_4 k) \left(\frac{-1 - \sqrt{3}i}{2} \right)^k$$

Ex. $d_n = 2d_{n-1} - d_{n-2} \Rightarrow d_n - 2d_{n-1} + d_{n-2} = 0$

CE $x^2 - 2x + 1 = 0$ $x = 1, x = 1$

$$d_n = (b_1 + b_2 n) (1)^n = b_1 + b_2 n$$

Ex. $a_n = 5a_{n-1} + 6a_{n-2}$ $n \geq 2$ $a_0 = a_1 = 3$

$a_n - 5a_{n-1} - 6a_{n-2} = 0$ CE $x^2 - 5x - 6 = 0$

$(x-6)(x+1) = 0$ $x = 6, -1$

$$a_n = b_1 \cdot 6^n + b_2 (-1)^n$$

$$a_0 = 3 = b_1 + b_2$$

$$a_1 = 3 = 6b_1 - b_2 \quad b_1 = \frac{6}{7} \quad b_2 = \frac{5}{7}$$

$$a_n = \frac{6}{7} (6)^n + \frac{5}{7} (-1)^n$$

Ex. $a_n + a_{n-1} + a_{n-2} = 0$

CE $x^2 + x + 1 = 0$

$x = \frac{-1 \pm \sqrt{3}i}{2}$

So $a_n = b_1 \left(\frac{-1 + \sqrt{3}i}{2} \right)^n + b_2 \left(\frac{-1 - \sqrt{3}i}{2} \right)^n$

Ex. $a_r - 6a_{r-1} + 12a_{r-2} - 8a_{r-3} = 0$

EE $x^3 - 6x^2 + 12x - 8 = 0$

$(x-2)^3 = 0$

roots = 2, 2, 2

$a_r = (b_1 + b_2 r + b_3 r^2) 2^r$

Ex. $a_{r+4} + 8a_{r+2} + 16a_r = 0$

$a_r = x^r$ $x^{r+4} + 8x^{r+2} + 16x^r = 0$

Div by x^r $x^4 + 8x^2 + 16 = 0$

$(x^2 + 4)^2 = 0$ $x = \pm 2i, \pm 2i$

$a_r = (b_1 + b_2 r)(2i)^r + (b_3 + b_4 r)(-2i)^r$

Ex. $a_r - 3a_{r-1} + 2a_{r-2} = 0$

$a_0 = 1$ $a_1 = 4$

CE $x^2 - 3x + 2 = 0$

$x = 1, 2$

$a_r = b_1 + b_2 2^r$

$a_0 = 1 = b_1 + b_2$

$a_1 = 4 = b_1 + 2b_2$

$b_2 = 3$ $b_1 = -2$

So $a_r = -2 + 3 \cdot 2^r$

Ex. $a_{n+1} - 1.5a_n = 0$ $n \geq 0$

CE $x - 1.5 = 0$

$x = 1.5$

$a_n = (1.5)^n$

Sol. of Non Homogeneous RR.

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$$a_n + C_1 a_{n-1} + C_2 a_{n-2} + \dots + C_k a_{n-k} = f(n).$$

- Homogeneous sol a_n^h with $f(n)=0$
- Particular sol a_n^p for $f(n)$ - Use Trial func
- Find coeff.
- $a_n = a_n^h + a_n^p$
- Find Const by using init cond.

$f(n)$	Trial func.
b^n (b is not root of CE)	$A b^n$
$P(n)$ Poly. of deg n .	$A_0 + A_1 n + \dots + A_m n^m$
b^n (b is root of CE of multiplicity s)	$A_0 n^s b^n$
$b^n P(n)$ b is not root of CE	$b^n (A_0 + A_1 n + \dots + A_m n^m)$
$b^n P(n)$ b is root of CE of mult's s	$b^n (A_0 + A_1 n + \dots + A_m n^m) n^s$
Constant	A

If $f(n)$ is a linear comb. of these then trial sol. is sum of corresp trial sol.

Ex. $y_k - y_{k-1} - 6y_{k-2} = -30$ $y_0 = 20$ $y_1 = -5$

for homo. sol $f(n)=0$ $y_k - y_{k-1} - 6y_{k-2} = 0$

CE ~~$y^2 - y - 6 = 0 = (y+3)(y-2)$~~

$(y-3)(y+2)=0$ $y = 3, -2$

homo. sol. $y_k^h = b_1 3^k + b_2 (-2)^k$

$f(n) = -30 = \text{Const.}$

Trial sol $y_k^p = A$

Substituting

$A - A - 6A = -30$ $A = 5$

so $y_k^p = 5$

$y_k = y_k^h + y_k^p = b_1 3^k + b_2 (-2)^k + 5$

$y_0 = 20 = b_1 + b_2 + 5$ $b_1 + b_2 = 15$

$y_1 = -5 = 3b_1 - 2b_2 + 5$ $3b_1 - 2b_2 = -10$

$b_1 = 4$ $b_2 = 11$

so $y_k = 4(3)^k + 11(-2)^k + 5$

Ex. $a_n - 6a_{n-1} + 8a_{n-2} = 3^n$ $a_0 = 3$ $a_1 = 7$

H.S. $a_n - 6a_{n-1} + 8a_{n-2} = 0$

CE $r^2 - 6r + 8 = 0 = (r-4)(r-2)$

$r = 2, 4$

$a_n^h = b_1 2^n + b_2 4^n$

P.S. $f(n) = 3^n$. 3 is not root

So Trial sol

$a_n^p = A \cdot 3^n$

Subst in $r r$.

$A 3^n - 6A 3^{n-1} + 8A 3^{n-2} = 3^n$

$A 3^{n-2} (9 - 18 + 8) = 3^n$

$-A \cdot 3^{n-2} = 9 \cdot 3^{n-2}$ $A = -9$

So $a_n^p = (-9) 3^n$

$a_n = a_n^h + a_n^p$

$a_n = b_1 2^n + b_2 4^n - 9 \cdot 3^n$

$a_0 = 3 = b_1 + b_2 - 9$ $b_1 + b_2 = 12$

$a_1 = 7 = 2b_1 + 4b_2 - 27$ $2b_1 + 4b_2 = 34$

$b_1 = 7$ $b_2 = 5$

$a_n = 7 \cdot 2^n + 5 \cdot 4^n - 9 \cdot 3^n$

$\frac{7}{2} - 3$

$\frac{1}{2}$

$\frac{16 - 28 + 12}{16}$

$\frac{16 - 28 + 12}{16}$

$\frac{16}{16}$

$\frac{-28 + 48}{16}$

En. $a_{r+2} - 5a_{r+1} + 6a_r = (r+1)^2$

HS $a_{r+2} - 5a_{r+1} + 6a_r = 0$

CE $r^2 - 5r + 6 = 0 = (r-3)(r-2) = 0$
 $r = 2, 3$

$a_n^h = b_1 2^n + b_2 3^n$

P.S. $f(n) = \text{Poly. of deg. } 2$

Trial Sol.

$a_n^p = A_0 + A_1 n + A_2 n^2$

Subst.

$$(\cancel{A_0 + A_1 n + A_2 n^2}) - 5[A_0 + A_1(n+1) + A_2(n+1)^2] + 6[A_0 + A_1(n+2) + A_2(n+2)^2] - 5[A_0 + A_1(n+1) + A_2(n+1)^2] + 6[A_0 + A_1 n + A_2 n^2] = r^2 + 2r + 1$$

~~$A_0 - 5A_0 + 6A_0 = 1$~~

$$A_0 + A_1 r + 2A_1 + r^2 A_2 + 2r A_2 + 4A_2 - 5A_0 - 5A_1 r - 5A_1 - 5A_2 - 10A_2 r - 5A_2 + 6A_0 + 6A_1 r + 6A_2 r^2 = r^2 + 2r + 1$$

Equating. $2A_0 - 3A_1 - A_2 = 1$ $A_2 = 1/2$

$-6A_2 + 2A_1 = 2 \Rightarrow A_1 = 5/2$

$2A_2 = 1$

$A_0 = 9/2$

$a_r^p = \frac{9}{2} + \frac{5}{2}r + \frac{1}{2}r^2$

$a_r = a_r^h + a_r^p = b_1 2^r + b_2 3^r + \frac{9}{2} + \frac{5}{2}r + \frac{1}{2}r^2$

~~$A_0 r^2 - 2r^2$~~

Ex. $a_r - 9a_{r-1} + 20a_{r-2} = 2 \cdot 5^r$

HS $a_r - 9a_{r-1} + 20a_{r-2} = 0$

CE $r^2 - 9r + 20 = 0 = (r-5)(r-4)$

$r = 4, 5$

$a_r^h = b_1 4^r + b_2 5^r$

~~Ans~~ Trial sol, as 5 is root of mult. 1

$a_r^p = \underline{A r 5^r}$

Substitute.

$A r 5^r - 9A(r-1)5^{r-1} + 20A(r-2)5^{r-2} = 2 \cdot 5^r$

$25A r 5^{r-2} - 45A(r-1)5^{r-2} + 20A(r-2)5^{r-2} = 50 \cdot 5^{r-2}$

$25A r - 45A(r-1) + 20A(r-2) = 50$

$25A r - 45A r + 45A + 20A r - 40A = 50$

$5A = 50 \quad A = 10$

$a_r^p = 10 r 5^r$

So

$a_r = a_r^h + a_r^p$

$= b_1 4^r + b_2 5^r + 10 r 5^r$

Ex. $a_r - 7a_{r-1} + 12a_{r-2} = r4^r$

HS $a_r - 7a_{r-1} + 12a_{r-2} = 0$

CE $r^2 - 7r + 12 = 0 = (r-3)(r-4)$
 $r = 3, 4$

$a_r^h = b_1 3^r + b_2 4^r$

for p.s.

4 is root of mult. 1.

so T.S. $= (A_0 + A_1 r) \cdot 4^r \cdot r' = (A_0 + A_1 r) r 4^r$

Subst.

$$[A_0 + A_1 r] r 4^r - 7[A_0 + A_1 (r-1)] (r-1) 4^{r-1} + 12[A_0 + A_1 (r-2)] (r-2) 4^{r-2} = r 4^r$$

$\Rightarrow$ ~~$r 4^r [A_0 + A_1 r - 7A_0 + 7A_1 (r-1) + 12A_0 - 24A_1 (r-2)] = r 4^r$~~

$$(A_0 + A_1 r) r 4^r - 7(A_0 r - A_0 + A_1 r^2 + A_1 r - 2A_1 r) 4^{r-1} + 12[A_0 r - 2A_0 + A_1 r^2 + 4A_1 - 4A_1 r] 4^{r-2} = r 4^r$$

$\Rightarrow$ ~~r~~ $\left[A_0 - \frac{7A_0}{4} + \frac{7A_1}{2} + \frac{12A_0}{16} - \frac{48A_1}{16} \right]$

$$+ r^2 \left[\frac{A_1}{4} - \frac{7A_1}{4} + \frac{12A_1}{16} \right] + \left[\frac{7A_0}{4} - \frac{7A_1}{4} - \frac{24A_0}{16} + \frac{48A_1}{16} \right] = r$$

$\Rightarrow r \left(\frac{A_1}{2} \right) + \left(\frac{1}{4} A_0 + \frac{5}{4} A_1 \right) = r$

Equating $\frac{A_1}{2} = 1$ $A_1 = 2$

$$\frac{1}{4} A_0 + \frac{5}{4} A_1 = 0$$

~~$$\frac{1}{4} A_0 + \frac{5}{4} A_1 = 0$$~~

$$\frac{1}{4} A_0 + \frac{10}{4} = 0$$

$$A_0 = -10$$

so

$$a_r^p = (-10 + 2r) r 4^r$$

$$a_r = b_1 3^r + b_2 4^r + (-10 + 2r) r 4^r$$

Ex. $S_r - 4S_{r-1} + 4S_{r-2} = 2^r + r2^r$

HS. $S_r - 4S_{r-1} + 4S_{r-2} = 0$

CE $\gamma^2 - 4\gamma + 4 = 0 = (\gamma - 2)^2 \quad \gamma = 2, 2$

$a_r^h = (b_1 + b_2 r) 2^r$

for PS. $f(r) = 2^r + r2^r = 2^r(r+1)$

Trial sol $(A_0 + A_1 r) r^2 2^r$

Subst.

$$r^2(A_0 + A_1 r) 2^r - 4(r-1)^2 [A_0 + A_1(r-1)] 2^{r-1} + 4(r-2)^2 [A_0 + A_1(r-2)] 2^{r-2} = 2^r + r2^r$$

$$4r^2(A_0 + A_1 r) 2^{r-2} - 8(r-1)^2 [A_0 + A_1(r-1)] 2^{r-2} + 4(r-2)^2 [A_0 + A_1(r-2)] 2^{r-2} = 4 \cdot 2^{r-2} + 4r 2^{r-2}$$

$$\Rightarrow 4r^2(A_0 + A_1 r) - 8(r^2 + 1 - 2r)(A_0 + A_1 r - A_1) + 4(r^2 + 4 - 4r)(A_0 + A_1 r - 2A_1) = 4 + 4r$$

$$\Rightarrow 4r^2 A_0 + 4r^3 A_1 - 8(r^2 A_0 + r^3 A_1 - r^2 A_1 + A_0 + A_1 r - A_1 - 2r A_0 - 2r^2 A_1 + 2r A_1) + 4(r^2 A_0 + A_1 r^3 - 2A_1 r^2 + 4A_0 + 4A_1 r - 8A_1 - 4A_0 r - 4A_1 r^2 + 8A_1 r) = 4 + 4r$$

Equating $-8A_0 + 8A_1 + 16A_0 - 32A_1 = 4 = 8A_0 - 24A_1$
 $2A_0 - 6A_1 = 1$

~~$$-8A_1 + 16A_0 - 16A_1 + 16A_1 + 32A_1 = 4$$~~

$$-8A_1 + 16A_0 - 16A_1 + 16A_1 - 16A_0 + 32A_1 = 4$$

$$24A_1 = 4 \quad A_1 = 1/6$$

$$A_0 = 1$$

So. $a_r^p = \left(1 + \frac{r}{6}\right) r^2 2^r$

$$a_r = (b_1 + b_2 r) 2^r + r^2 \left(\frac{r}{6} + 1\right) 2^r$$

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En.

$$a_r - 6a_{r-1} + 8a_{r-2} = r4^r$$

$$a_0 = 8 \quad a_1 = 22$$

$$\text{HS} \quad a_r - 6a_{r-1} + 8a_{r-2} = 0$$

$$\text{CE} \quad r^2 - 6r + 8 = 0 = (r-4)(r-2)$$

$$r = 4, 2$$

~~$$a_r^h = b_1 4^r + b_2 2^r$$~~

$$a_r^h = b_1 4^r + b_2 2^r$$

~~$$a_r^h = (a_1 + a_2)r + 4^r$$~~

$$\text{Trial sol.} \quad f(r) = r4^r \quad (A_0 + A_1 r)r4^r$$

subst.

$$(A_0 + A_1 r)r4^r - 6[A_0 + A_1(r-1)](r-1)4^{r-1} + 8[A_0 + A_1(r-2)](r-2)4^{r-2} = r4^r$$

$$16r(A_0 + A_1 r)4^{r-2} - 24[A_0 + A_1(r-1)](r-1)4^{r-2} + 8[A_0 + A_1(r-2)](r-2)4^{r-2} = 16r4^{r-2}$$

$$= 16r4^{r-2}$$

$$16rA_0 + 16A_1 r^2 - 24(A_0 r - A_0 + A_1 r^2 + A_1 - 2rA_1) + 8(A_0 r - 2A_0 + A_1 r^2 - 4rA_1 + 4A_0) = 16r$$

$$16rA_0 + 16A_1 r^2 - 24A_0 r + 24A_0 - 24A_1 r^2 - 24A_1 + 48rA_1 + 8A_0 r - 16A_0 + 8A_1 r^2 + 32A_1 - 32rA_1 = 16r$$

$$\Rightarrow 16A_0 - 24A_0 + 48A_1 - 32A_1 = 16$$

~~$$16A_0 + 16A_1 = 16$$~~

Company.

$$\Rightarrow r^0 \quad 24A_0 - 24A_1 - 16A_0 + 32A_1 = 0 \quad A_0 + A_1 = 0$$

~~$$r^1 \quad 16A_0 - 24A_0 + 48A_1 - 32A_1 = 16$$~~

$$r^2 \quad 16A_1 - 24A_1 + 8A_1 = 0$$

$$r \quad 16A_0 - 24A_0 + 48A_1 + 8A_0 - 32A_1 = 16$$

$$A_1 = 1$$

$$A_0 = -1$$

$$\text{So} \quad a_r^p = (-1 + r)r4^r$$

So

$$a_r = a_r^h + a_r^p = b_1 4^r + b_2 2^r + (-1 + r)r4^r$$

$$a_0 = 8$$

$$8 = b_1 + b_2$$

$$a_1 = 22$$

$$22 = 4b_1 + 2b_2$$

$$\Rightarrow b_1 = 5 \quad b_2 = 3$$

$$\text{So} \quad a_r = 5 \cdot 4^r + 3 \cdot 2^r + (-1 + r)r4^r$$

Ys. $f(k) = 9k + 1$
~~Trial sol = $(A_0 + A_1 k)$~~

Ex. $y_{n+2} - y_{n+1} - y_n = n^2$

HS CE $r^2 - 1 + 1 = 0$ $r = \frac{1 \pm \sqrt{5}}{2}$

$y_n^h = b_1 \left(\frac{1+\sqrt{5}}{2} \right)^n + b_2 \left(\frac{1-\sqrt{5}}{2} \right)^n$

Ps $f(n) = n^2$ $a_n^p = A_0 + A_1 n + A_2 n^2$

Subst.

$A_0 + A_1(n+2) + A_2(n+2)^2 - A_0 - A_1(n+1) - A_2(n+1)^2 - A_0 - A_1 n - A_2 n^2 = n^2$

$A_0 + A_1 n + 2A_1 + A_2 n^2 + 4A_2 + 4nA_2 - A_0 - A_1 n - A_1 - A_2 n^2 - A_2 - 2nA_2 - A_0 - A_1 n - A_2 n^2 = n^2$

Compare.

$A_2 - A_2 - A_2 = 1$ $A_2 = -1$

$A_1 + 4A_2 - A_1 - 2A_2 - A_1 = 0$ $2A_2 - A_1 = 0$ $A_1 = 2A_2 = -2$

$A_0 + 2A_1 + 4A_2 - A_0 - A_1 - A_2 - A_0 = 0$
 $-4 - 4 + 2 + 1 - A_0 = 0$ $-5 - A_0 = 0$ $A_0 = -5$
 $y_n^p = -5 - 2n - n^2$

So

$y_n = b_1 \left(\frac{1+\sqrt{5}}{2} \right)^n + b_2 \left(\frac{1-\sqrt{5}}{2} \right)^n - 5 - 2n - n^2$

GENERATING FUNCTIONS

①

Def: If $\{a_0, a_1, a_2, \dots, a_n, \dots\}$ is a sequence of real or complex numbers, then the power series given by:

$$A(z) = \sum_{r=0}^{\infty} a_r z^r = a_0 + a_1 z + a_2 z^2 + \dots + a_n z^n + \dots$$

is called the generating function for the given seqⁿ.

It is evident from here that $(n+1)^{\text{th}}$ term in this series $= a_n = \text{coefficient of } z^n \text{ in } A(z)$.

Determination of generating function for a given seqⁿ.

Ex.1. Find the generating function.

(a) $1, -1, 1, -1, 1, -1, \dots$

Sol. (a) The required generating function is given by

$$A(z) = \sum_{r=0}^{\infty} a_r z^r = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \dots$$

$$= 1 + (-1)z + 1 \cdot z^2 + (-1)z^3 + \dots$$

$$= 1 - z + z^2 - z^3 + \dots$$

$$= \frac{1}{1 - (-z)} \quad [\text{by summing the infinite G.P.}]$$

$$= \frac{1}{1+z} \quad \text{or } (1+z)^{-1}$$

(b) $\{a, a, \dots\}$

Sol. Here $a_r = a$ for $r = 0, 1, 2, \dots$

Hence the req generating function is

$$A(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \dots$$

$$= a + az + az^2 + az^3 + \dots$$

$$= a(1 + z + z^2 + \dots)$$

$$= a \left(\frac{1}{1-z} \right) \quad (\text{Summing the infinite G.P.})$$

$$= a(1-z)^{-1}$$

②

(c) $b_n = ar^n$, $n \geq 0$ i.e., $a, ar, ar^2, \dots$

Sol. $A(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + \dots$

Here $a_0 = a$, $a_1 = ar$, $a_2 = ar^2, \dots$

Therefore, $A(z) = a + ar.z + ar^2 z^2 + ar^3 z^3 + \dots$

$$= \frac{a}{(1-rz)} \quad (\text{on summing infinite G.P.})$$

Ex. 2. Find the generating fⁿs which generated by the sequence

(a) $1, 2, 4, 8, 16, 32, \dots$

Sol (a) $A(z) = 1 + 2z + 4z^2 + 8z^3 + 16z^4 + \dots$

$$= 1 + 2z + (2z)^2 + (2z)^3 + (2z)^4 + \dots$$

$$= \frac{1}{1-2z}$$

(b) $1, \frac{2}{1!}, \frac{4}{2!}, \frac{8}{3!}, \frac{16}{4!}, \dots$

Sol (b) $A(z) = 1.z^0 + \frac{2}{1!} z + \frac{4}{2!} z^2 + \frac{8}{3!} z^3 + \dots$

$$= 1 + \frac{2z}{1!} + \frac{(2z)^2}{2!} + \frac{(2z)^3}{3!} + \dots = e^{2z}$$

1.1. If the sequence is finite say $a_0, a_1, a_2, \dots, a_n$ then the corresponding finite generating function is defined as

$$A(z) = a_0 + a_1 z + a_2 z^2 + \dots + a_n z^n$$

$$= \sum_{r=0}^n a_r z^r$$

Ex. 3. Find the generating function for the finite seqⁿ

$1, 2, 2^2, 2^3, 2^4$

Sol. $A(z) = 1.z^0 + 2.z^1 + 2^2.z^2 + 2^3.z^3 + 2^4.z^4$

$$= 1 + 2z + (2z)^2 + (2z)^3 + (2z)^4$$

$$= \frac{1 \cdot ((2z)^5 - 1)}{2z - 1}$$

(on summing the G.P.)

Ex. 4 Find G.F. for the sequence $a_n = 5^n$. ($n \geq 0$) and hence calculate the G.F. of $7 \cdot 5^n$.

Sol. The given sequence $a_0, a_1, a_2, a_3, \dots$

can be written as $1, 5, 5^2, 5^3, \dots$

Therefore the G.F. of this sequence is given by

$$A(z) = 1 \cdot z^0 + 5 \cdot z^1 + 5^2 \cdot z^2 + 5^3 \cdot z^3 + \dots$$

$$= \frac{1}{1-5z} \quad (\text{on summing the G.P.}) \quad \dots (1)$$

Again G.F. of the seqⁿ $b_n = 7 \cdot 5^n$

$$= 7 \cdot (\text{G.F. of seqⁿ } a_n = 5^n)$$

$$= \frac{7}{1-5z} \quad \text{from (1)}$$

Some rules regarding Binomial Theorem and Generating Functions

Rule 1. $(1+z)^n = \sum_{r=0}^{\infty} {}^nC_r \cdot z^r$ if $n \geq 0$

Rule 2. Coefficient of z^r in the expansion of $(1-az)^{-n}$

$$= {}^{n-1+r}C_r \cdot a^r$$

Rule 3. If $A(z)$ is the generating function for the seqⁿ $\{a_n\}_{n \geq 0}$ then generating function $S(z)$ for the seqⁿ $\{S_n\}$ of its partial sums $\left[S_n = \sum_{k=0}^n a_k, (n \geq 0) \right]$ is given by

$$S(z) = A(z)/(1-z).$$

Rule 4. If $A(z)$ is the generating function for the seqⁿ $\{a_n\}_{n \geq 0}$ and $B(z)$ is the generating function for the seqⁿ $\{b_n\}_{n \geq 0}$, then the generating function $\{c_n\}_{n \geq 0}$

$$c_n = \sum_{k=0}^n a_k \cdot b_{n-k} \text{ is given by } C(z) = A(z) \times B(z)$$

④

Rule 5: If $A(z)$ is the generating function for the seqⁿ $\{a_n\}_{n \geq 0}$, then the generating function for the seqⁿ $\{b_n\}_{n \geq 0}$, where $b_n = a_n - a_{n-1}$ for $n \geq 1$ and $b_0 = a_0$ is given by $B(z) = (1-z) \cdot A(z)$.

NOTE: The coefficient of z^n in the expansion of $(1-az)^{-1} = (a)^n$ and the coefficient of z^n in the expansion of $(1+az)^{-1} = (-a)^n$.

Ex. 5. If the generating function of a sequence $a_0, a_1, a_2, \dots$ is given by $A(z) = 2/(1-4z^2)$, find a_n the $(n+1)^{\text{th}}$ term of the seqⁿ.

Sol.
$$A(z) = \frac{2}{1-4z^2} = \frac{2}{(1-2z)(1+2z)}$$

$$= \frac{1}{1-2z} + \frac{1}{1+2z} \quad \dots (1) \text{ (by partial fract)}$$

Now the $(n+1)^{\text{th}}$ term $a_n =$ coefficient of z^n in the expansion of (1)

$$= 2^n + (-2)^n$$

Ex. 6. Find the discrete numeric function corresponding to the generating function $\frac{1}{5-6z+z^2}$

Sol.
$$\text{G.F.} = \frac{1}{5-6z+z^2}$$

$$= \frac{1}{4(1-z)} - \frac{1}{4(5-z)}, \quad (\text{after partial fract})$$

Therefore the general term of the numeric function corresponding to G.F. is

$$a_r = \text{Coefficient of } z^r \text{ in } \frac{1}{4(1-z)} - \frac{1}{4(5-z)}$$

$$\begin{aligned} &= \text{Coefficient of } z^r \text{ in } \left[\frac{1}{4} (1-z)^{-1} - \frac{1}{4} \cdot \frac{1}{5} \left(1 - \frac{z}{5}\right)^{-1} \right] \\ &= \frac{1}{4} \text{ Coefficient of } z^r \text{ in } (1-z)^{-1} - \frac{1}{20} \cdot \text{Coef. of } z^r \text{ in } \left(1 - \frac{z}{5}\right)^{-1} \\ &= \frac{1}{4} \cdot {}^{1-1+r}C_r \cdot (1)^r \left(\frac{1}{5}\right)^r - \frac{1}{20} {}^{1-1+r}C_r \cdot \left(\frac{1}{5}\right)^r \\ &= \frac{1}{4} - \frac{1}{20} \left(\frac{1}{5}\right)^r \\ &= \frac{1}{4} \left[1 - \left(\frac{1}{5}\right)^{r+1} \right], \quad r \geq 0 \end{aligned}$$

Solution of Recurrence Relations with the help of G.F.

Ex.1 Solve the recurrence relation $u_n = u_{n-1} + 2u_{n-2}, n \geq 2,$
 $u_0 = 3, u_1 = 7.$

Sol. The given recurrence relation is
 $u_n = u_{n-1} + 2u_{n-2}, \quad n \geq 2 \quad \dots (1)$

Multiplying both the sides of (1) by $z^n,$
 $u_n \cdot z^n = u_{n-1} z^n + 2u_{n-2} z^n \quad \dots (2)$

Summing both the sides of (2) from $n=2$ to $\infty,$

$$\sum_{n=2}^{\infty} u_n z^n = \sum_{n=2}^{\infty} u_{n-1} z^n + 2 \sum_{n=2}^{\infty} u_{n-2} z^n$$

$$\text{or } \sum_{n=2}^{\infty} u_n z^n = z \sum_{n=2}^{\infty} u_{n-1} z^{n-1} + 2z^2 \sum_{n=2}^{\infty} u_{n-2} z^{n-2} \quad \dots (3)$$

(Keeping the exponent of z equal to the suffix of u inside the summation sign).

Now we shall express each term of (3) in terms of G.F.

$A(z) = u_0 + u_1 z + u_2 z^2 + \dots$ as explained below.

The term on L.H.S. of (3) is

$$\begin{aligned} \sum_{n=2}^{\infty} u_n z^n &= u_2 z^2 + u_3 z^3 + \dots \\ &= (u_0 + u_1 z + u_2 z^2 + u_3 z^3 + \dots) - (u_0 + u_1 z) \\ &= A(z) - u_0 - u_1 z \quad \dots (4) \end{aligned}$$

⑥

The first term of R.H.S. in (3) is

$$\begin{aligned} z \sum_{n=2}^{\infty} u_{n-1} z^{n-1} &= z [u_1 z + u_2 z^2 + u_3 z^3 + \dots] \\ &= z [A(z) - u_0] \\ &= z A(z) - u_0 z \end{aligned} \quad \dots (5)$$

The second term of R.H.S. in (3) is

$$\begin{aligned} 2z^2 \sum_{n=2}^{\infty} u_{n-2} z^{n-2} &= 2z^2 [u_0 z^0 + u_1 z + u_2 z^2 + \dots] \\ &= 2z^2 A(z) \end{aligned} \quad \dots (6)$$

Putting values of different terms from (4), (5) & (6) in (3),

$$A(z) - u_0 - u_1 z = z A(z) - u_0 z + 2z^2 A(z)$$

$$A(z) (1 - z - 2z^2) = u_0 + u_1 z - u_0 z$$

$$A(z) = \frac{u_0 + u_1 z - u_0 z}{1 - z - 2z^2} = \frac{3 + 7z - 3z}{1 - z - 2z^2} \quad \dots (7)$$

$$= \frac{3 + 4z}{(1 - 2z)(1 + z)} = \frac{10}{3(1 - 2z)} - \frac{1}{3(1 + z)} \quad (\text{by Partial fraction})$$

Therefore the value of u_n = Coefficient of z^n in the R.H.S. of (7)

$$= \frac{10}{3} (2^n) - \frac{1}{3} (-1)^n \quad \underline{\text{Ans.}}$$

Ex. 2. Solve the recurrence relation

$$u_n = 6u_{n-1} + 2^{n-1}, \quad u_0 = 1$$

$$\text{Sol. } u_n = 6u_{n-1} + 2^{n-1} \quad \dots (1)$$

Multiplying both the sides of (1) by z^n and summing up from $n=1$ to $n=\infty$,

$$\sum_{n=1}^{\infty} u_n z^n = 6 \sum_{n=1}^{\infty} u_{n-1} z^n + \sum_{n=1}^{\infty} 2^{n-1} z^n$$

$$\sum_{n=1}^{\infty} u_n z^n = 6z \sum_{n=1}^{\infty} u_{n-1} z^{n-1} + z \sum_{n=1}^{\infty} 2^{n-1} z^{n-1} \quad \dots (2)$$

Now the term on the L.H.S. of (2) is

$$\sum_{n=1}^{\infty} u_n z^n = u_1 z + u_2 z^2 + \dots = A(z) - u_0 \quad \dots (3)$$

Again first term on R.H.S. of (2) is

$$6z \sum_{n=1}^{\infty} u_{n-1} z^{n-1} = 6z [u_0 + u_1 z + u_2 z^2 + \dots] \\ = 6z \cdot A(z) \quad \text{--- (4)}$$

Second term on R.H.S. of (2) is

$$z \sum_{n=1}^{\infty} 2^{n-1} \cdot z^{n-1} = z [1 + 2z + 2^2 z^2 + \dots] \\ = \frac{z}{1-2z} \quad (\text{on summing infinite G.P.}) \text{--- (5)}$$

Putting values of all the term from (3), (4) & (5) in (2)

$$A(z) - u_0 = 6z A(z) + \frac{z}{1-2z}$$

$$A(z) (1-6z) = u_0 + \frac{z}{1-2z}$$

$$A(z) = \frac{u_0}{1-6z} + \frac{z}{(1-2z)(1-6z)} = \frac{1}{1-6z} + \frac{z}{(1-2z)(1-6z)}$$

$$= \frac{1-z}{(1-2z)(1-6z)} = \frac{5}{4(1-6z)} - \frac{1}{4(1-2z)} \quad \text{--- (6)}$$

Therefore, $u_n = \text{Coefficient of } z^n \text{ in R.H.S. of (6)}$

$$= \frac{5}{4} (6^n) - \frac{1}{4} (2^n).$$

$$= \frac{1}{4} [5 \cdot 6^n - 2^n]. \quad \underline{\text{Ans.}}$$

Generating function

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①

①

General Term.

GF $G(x)$

1

$$1/(1-x)$$

✓

$k+1$

$$1/(1-x)^2$$

✓

k

$$x/(1-x)^2$$

✓

$k(k+1)$

$$2x/(1-x)^3$$

$(k+1)(k+2)$

$$2/(1-x)^3$$

a^k

$$1/(1-ax)$$

✓

The GF for the sequence $a_0, a_1, \dots, a_k, \dots$ of real no. is a series

$$G(x) = a_0 + a_1x + a_2x^2 + \dots + a_kx^k + \dots = \sum_{k=0}^{\infty} a_k x^k.$$

Ex.

1, 2, 4, 8, ...

$$a_0 = 1, a_1 = 2, a_2 = 4, \dots$$

$$G(x) = 1 + 2x + 4x^2 + \dots = \sum_{n=0}^{\infty} 2^n x^n.$$

$$2x G(x) = 2x + 4x^2 + 8x^3 + \dots$$

$$G(x) - 2x G(x) = 1$$

$$G(x) = \frac{1}{1-2x}.$$

Ex.

1, -1, 1, -1, ...

$$a_0 = 1, a_1 = -1, \dots$$

$$G(x) = 1 - x + x^2 - x^3 + \dots = \sum_{n=0}^{\infty} (-1)^n x^n.$$

$$x G(x) = x - x^2 + x^3 - x^4 + \dots$$

$$G(x) + x G(x) = 1$$

$$G(x) = \frac{1}{1+x}$$

Ex.

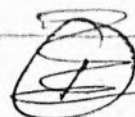
1, 1, 1, 1, ...

$$G(x) = 1 + x + x^2 + x^3 + \dots = \sum_{n=0}^{\infty} 1^n x^n$$

$$x G(x) = x + x^2 + x^3 + \dots$$

$$G(x) - x G(x) = 1$$

$$G(x) = \frac{1}{1-x}.$$



Ex. $0, 1, 2, 3, 4, \dots$

$$G(x) = 0x + 1x + 2x^2 + 3x^3 + \dots$$

$$xG(x) = x^2 + 2x^3 + 3x^4 + 4x^5 + \dots$$

$$G(x) - xG(x) = \cancel{0x} + x + x^2 + x^3 + \dots = \cancel{0x}$$

$$G(x) - xG(x) = x(1 + x + x^2 + \dots) = x \cdot \sum_{n=0}^{\infty} 1^n x^n$$

$$G(x)(1-x) = x \cdot \frac{1}{1-x}$$

$$G(x) = \frac{x}{(1-x)^2}$$

Ex. $1, 2, 3, 4, \dots$

$$G(x) = 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$xG(x) = x + 2x^2 + 3x^3 + 4x^4 + \dots$$

$$G(x) - xG(x) = 1 + x + x^2 + \dots = \sum_{n=0}^{\infty} 1^n x^n$$

$$G(x)(1-x) = \frac{1}{1-x}$$

$$G(x) = \frac{1}{(1-x)^2}$$

Ex. $1, 1, 0, 1, 1, 1, \dots$

$$G(x) = 1 + 1x + 0x^2 + 1x^3 + 1x^4 + \dots$$

$$= 1 + x + x^3 + x^4 + \dots$$

$$= \cancel{0x^2} + 1 + x + x^2 + x^3 + \dots - x^2$$

$$= \frac{1}{1-x} - x^2$$

Ex. $0, 0, 1, 1, 1, \dots$

$$G(x) = 0 + 0x + 1x^2 + 1x^3 + \dots$$

$$= x^2(1 + x + \dots)$$

$$= \frac{x^2}{1-x}$$

Ex. $1, a, a^2, \dots$

$$G(x) = 1 + ax + a^2x^2 + \dots = \sum_{n=0}^{\infty} a^n x^n$$

$$ax G(x) = ax + a^2x^2 + a^3x^3 + \dots$$

$$G(x) - ax G(x) = 1$$

$$G(x) = \frac{1}{1-ax}$$

Ex. Nam func for GF

$$\rightarrow 9/(1-5x)$$

$$G(x) = \sum_{n=0}^{\infty} a_n x^n = 9(1-5x)^{-1}$$

$$= 9 \sum_{n=0}^{\infty} 5^n x^n = 9 \cdot 5^n \quad n \geq 0$$

$$\rightarrow \frac{10}{1-x} + \frac{12}{2-x} = \frac{10}{1-x} + \frac{6}{1-x/2}$$

$$\sum_{n=0}^{\infty} a_n x^n = \frac{10}{1-x} + \frac{6}{1-x/2}$$

$$= 10 \sum_{n=0}^{\infty} 1^n x^n + 6 \cdot \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n x^n$$

$$= 10 \cdot 1^n + 6 \cdot \left(\frac{1}{2}\right)^n = 10 + 6 \cdot \left(\frac{1}{2}\right)^n \quad n \geq 0$$

$$\rightarrow \frac{6-29x}{1-11x+30x^2} = \frac{6-29x}{(1-6x)(1-5x)}$$

using Partial fractions.

$$= \frac{7}{1-6x} - \frac{1}{1-5x} = 7 \sum_{n=0}^{\infty} 6^n x^n - \sum_{n=0}^{\infty} 5^n x^n$$

$$= 7 \cdot 6^n - 5^n \quad n \geq 0$$

Method of Partial fraction.

$$\frac{6-29x}{(1-6x)(1-5x)} = \frac{A}{1-6x} + \frac{B}{1-5x}$$

$$6-29x = A(1-5x) + B(1-6x)$$

$$x = \frac{1}{5} \quad 6 - \frac{29}{5} = A(1 - 5 \cdot \frac{1}{5}) + B(1 - \frac{6}{5}) \Rightarrow B = -1$$

$$x = \frac{1}{6} \quad 6 - \frac{29}{6} = A(1 - 5 \cdot \frac{1}{6}) + B(1 - 1) \Rightarrow A = 7$$

$$\text{So } \frac{7}{1-6x} - \frac{1}{1-5x}$$

Ex. $1, \frac{2}{3}, \frac{3}{9}, \frac{4}{27}, \dots$

$$G(x) = 1 + \frac{2}{3}x + \frac{3}{9}x^2 + \frac{4}{27}x^3 + \dots + \frac{n+1}{3^n}x^n + \dots$$

$$\frac{1}{3}x G(x) = \frac{1}{3}x + \frac{2}{9}x^2 + \frac{3}{27}x^3 + \dots$$

$$G(x) - \frac{1}{3}x G(x) = 1 + \frac{1}{3}x + \frac{1}{9}x^2 + \frac{1}{27}x^3 + \dots$$

$$G(x)(1 - \frac{1}{3}x) = \frac{1}{1 - \frac{1}{3}x}$$

$$G(x) = \frac{1}{(1 - \frac{1}{3}x)^2} = \frac{9}{(3-x)^2}$$

Ex $1, 0, -1, 0, 1, 0, -1, 0, 1$

$$G(x) = 1 + 0 \cdot x - 1 \cdot x^2 + 0 \cdot x^3 + 1 \cdot x^4 + 0 \cdot x^5 - 1 \cdot x^6 + \dots$$

$$= 1 - x^2 + x^4 - x^6 + \dots$$

$$= 1 + (-x^2) + (-x^2)^2 + (-x^2)^3 + \dots$$

$$= \frac{1}{1+x^2}$$

Ex. $3, -3, 3, -3, \dots$

$$G(x) = 3 - 3x + 3x^2 - 3x^3 + \dots$$

$$= 3[1 - x + x^2 - x^3 + \dots]$$

$$= 3$$

$$\frac{3}{1+x}$$

method for finding for both

$$3 + \frac{A}{x^2-1} = \frac{KMS-2}{x^2-1}$$

$$x^2-1 = x^2-1 \quad (x^2-1)(x^2-1)$$

$$(x^2-1)3 = (x^2-1)A = x^2MS-2$$

Solving RR using GF.

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②

⑤

Ex.

$$a_{r+2} - 3a_{r+1} + 2a_r = 0$$

$$a_0 = 2 \quad a_1 = 3$$

Let $G(t) = \sum_{r=0}^{\infty} a_r t^r$ be GF of sequence $\{a_r\}$

multiply the RR with t^r and sum from 0 to ∞ .

$$\sum_{r=0}^{\infty} a_{r+2} t^r - 3 \sum_{r=0}^{\infty} a_{r+1} t^r + 2 \sum_{r=0}^{\infty} a_r t^r = 0$$

$$\Rightarrow (a_2 + a_3 t + a_4 t^2 + \dots) - 3(a_1 + a_2 t + a_3 t^2 + \dots) + 2(a_0 + a_1 t + \dots) = 0$$

✶

$$G(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + \dots$$

$$\Rightarrow \frac{G(t) - a_0 - a_1 t}{t^2} - 3 \frac{(G(t) - a_0)}{t} + 2 G(t) = 0$$

$$a_0 = 2 \quad a_1 = 3$$

$$\frac{G(t) - 2 - 3t}{t^2} - 3 \frac{(G(t) - 2)}{t} + 2G(t) = 0$$

$$\frac{G(t) - 2 - 3t}{t^2} - \frac{3G(t) + 6}{t} + 2G(t) = 0$$

$$G(t) \left(\frac{1}{t^2} - \frac{3}{t} + 2 \right) = \frac{2 + 3t - 6}{t}$$

$$G(t) \left(\frac{1 - 3t + 2t^2}{t^2} \right) = \frac{2 + 3t - 6}{t}$$

$$G(t) (1 - 3t + 2t^2) = (2 + 3t - 6)t$$

$$G(t) = \frac{2 - 3t}{1 - 3t + 2t^2} = \frac{2 - 3t}{(1-t)(1-2t)}$$

Using p.f.

$$\frac{2 - 3t}{(1-t)(1-2t)} = \frac{A}{1-t} + \frac{B}{1-2t} = A(1-2t) + B(1-t)$$

$$t=1 \quad 2-3 = A(1-2) \quad -1 = -A \quad A=1$$

$$t=1/2 \quad 2-3/2 = B(1-1/2) \quad 1/2 = B/2 \quad B=1$$

$$G(t) = \frac{1}{1-t} + \frac{1}{1-2t}$$

$$\sum_{r=0}^{\infty} a_r t^r = \sum_{r=0}^{\infty} t^r + \sum_{r=0}^{\infty} 2^r t^r$$

$$a_r = 1 + 2^r \quad r \geq 0$$

(6)

as smallest
needs to

Ex. $a_r - 7a_{r-1} + 10a_{r-2} = 0$ $a_0 = 3$ $a_1 = 3$.

Let $G(t) = \sum_{r=0}^{\infty} a_r t^r$ be GF of $\{a_r\}$.
multiply RR with t^r & sum from 2 to ∞ .

$$\sum_{r=2}^{\infty} a_r t^r - 7 \sum_{r=2}^{\infty} a_{r-1} t^r + 10 \sum_{r=2}^{\infty} a_{r-2} t^r = 0$$

$$= (a_2 t^2 + a_3 t^3 + \dots) - 7(a_1 t^2 + a_2 t^3 + \dots) + 10(a_0 t^2 + a_1 t^3 + \dots)$$

$$= G(t)$$

$$G(t) = a_0 + a_1 t + a_2 t^2 + \dots$$

$$= (G(t) - a_0 - a_1 t) - 7(tG(t) - t a_0) + 10(t^2 G(t) - t^2 a_0) = 0$$

$$G(t) - a_0 - a_1 t - 7tG(t) + 7ta_0 + 10t^2 G(t) - 10t^2 a_0 = 0$$

$$G(t)(1 - 7t + 10t^2) - a_0 - a_1 t + 7ta_0 - 10t^2 a_0 = 0$$

$$a_0 = a_1 = 3$$

$$G(t)(1 - 7t + 10t^2) - 3 - 3t + 21t - 30t^2 = 0$$

$$G(t)(1 - 7t + 10t^2) - 3 + 18t - 30t^2 = 0$$

$$G(t) = \frac{3 - 18t + 30t^2}{1 - 7t + 10t^2} = \frac{3 - 18t}{(1 - 2t)(1 - 5t)}$$

using pf

$$\frac{3 - 18t}{(1 - 2t)(1 - 5t)} = \frac{A}{1 - 2t} + \frac{B}{1 - 5t} = A(1 - 5t) + B(1 - 2t)$$

$$t = 1/2 \quad \frac{3 - 18 \cdot 1/2}{1 - 2 \cdot 1/2} = A(1 - 5 \cdot 1/2) \quad -12 = A \cdot -3/2 \quad A = 8$$

$$t = 1/5 \quad \frac{3 - 18 \cdot 1/5}{1 - 2 \cdot 1/5} = B(1 - 5 \cdot 1/5) \quad -3/5 = B \cdot 0 \quad B = -1$$

So

$$G(t) = \frac{8}{1 - 2t} - \frac{1}{1 - 5t}$$

$$\sum_{r=0}^{\infty} a_r t^r = 8 \sum_{r=0}^{\infty} 2^r t^r - \sum_{r=0}^{\infty} 5^r t^r$$

$$a_r = 8 \cdot 2^r - 5^r$$

Ex. $a_{r+2} - 2a_{r+1} + a_r = 2^r$ $a_0 = 2$ $a_1 = 1$.

$G(t) = \sum_0^\infty a_r t^r$ multiply RR with t^r & sum from 0 to ∞ .

$$\sum_0^\infty a_{r+2} t^r - 2 \sum_0^\infty a_{r+1} t^r + \sum_0^\infty a_r t^r = \sum_0^\infty 2^r t^r$$

OR

$$\Rightarrow (a_2 + a_3 t + a_4 t^2 + \dots) - 2(a_1 + a_2 t + a_3 t^2 + \dots) + (a_0 + a_1 t + \dots) = \frac{1}{1-2t}$$

$$G(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + \dots$$

$$\frac{G(t) - a_0 - a_1 t}{t^2} - 2 \frac{(G(t) - a_0) + G(t)}{t} = \frac{1}{1-2t}$$

~~$G(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + \dots$~~

$$G(t) - a_0 - a_1 t - 2t(G(t) - a_0) + t^2 G(t) = \frac{t^2}{1-2t}$$

$$a_0 = 2 \quad a_1 = 1$$

$$G(t) - 2 - t - 2t(G(t) - 2) + t^2 G(t) = \frac{t^2}{1-2t}$$

$$G(t)(1-2t+t^2) - 2 - t + 4t = \frac{t^2}{1-2t}$$

$$G(t)(1-t)(1-t) - 2 + 3t = \frac{t^2}{1-2t}$$

$$G(t) = \left(\frac{2-3t+t^2}{1-2t} \right) \cdot \frac{1}{1-t^2}$$

$$G(t) = \frac{2}{1-t^2} - \frac{3t}{1-t^2} + \frac{t^2}{(1-2t)(1-t^2)}$$

Using PF.

$$\frac{t^2}{(1-2t)(1-t^2)} = \frac{A}{1-2t} + \frac{B}{1-t^2} \quad t^2 = A(1-t^2) + B(1-2t)$$

$$t=1 \quad 1 = B(1-2) \Rightarrow B = -1$$

$$t = 1/2 \quad 1/4 = A(1-1/4) \quad A = 3$$

$$G(t) = \frac{2}{1-t^2} - \frac{3t}{1-t^2} + \frac{3}{1-2t} - \frac{1}{1-t^2}$$

$$G(t) = \frac{1}{1-t^2} - \frac{3t}{1-t^2} + \frac{3}{1-2t}$$

$$\sum_0^\infty a_r t^r = \sum_0^\infty (r+1)t^r - 3 \sum_0^\infty r t^r + 3 \sum_0^\infty 2^r t^r$$

$$a_r = r+1 - 3r + 3 \cdot 2^r$$

$$a_r = \underline{\underline{1 - 2r + 3 \cdot 2^r}}$$

Q. $a_{n+2} - 5a_{n+1} + 6a_n = 2$ $a_0 = 1$ $a_1 = 2$.
 $G(t) = \sum_{n=0}^{\infty} a_n t^n$ multiply by t^r with t^r & sum from 0 to ∞
 $\sum_{n=0}^{\infty} a_{n+2} t^n = 5 \sum_{n=0}^{\infty} a_{n+1} t^n + 6 \sum_{n=0}^{\infty} a_n t^n = 2 \sum_{n=0}^{\infty} t^n$

~~$(a_2 + a_3 t + a_4 t^2 + \dots) - 5(a_1 + a_2 t + \dots) + 6(a_0 + a_1 t + \dots) = \frac{2}{1-t}$~~
 $G(t) - a_0 - a_1 t - 5(G(t) - a_0) + 6G(t) = \frac{2}{1-t}$

$G(t) - a_0 - a_1 t - 5t(G(t) - a_0) + 6t^2 G(t) = \frac{2t^2}{1-t}$

$a_0 = 1$ $a_1 = 2$

$G(t) - 1 - 2t - 5t(G(t) - 1) + 6t^2 G(t) = \frac{2t^2}{1-t}$

$G(t)(1 - 5t + 6t^2) - 1 + 3t = \frac{2t^2}{1-t}$

$G(t) = \left(\frac{1 - 3t + 2t^2}{1-t} \right) \frac{1}{(1-2t)(1-3t)}$

$= \frac{(1-3t)(1-t) + 2t^2}{(1-t)(1-2t)(1-3t)} = \frac{1-4t+5t^2}{(1-t)(1-2t)(1-3t)}$

Using PF

$\frac{5t^2 - 4t + 1}{(1-t)(1-2t)(1-3t)} = \frac{A}{(1-t)} + \frac{B}{(1-2t)} + \frac{C}{(1-3t)}$

$5t^2 - 4t + 1 = A(1-2t)(1-3t) + B(1-t)(1-3t) + C(1-t)(1-2t)$

$t=1$ $5-4+1=2= A(1-2)(1-3) = 2A$ $A=1$

$t=1/2$ $\frac{5}{4} - \frac{4}{2} + 1 = \frac{1}{4} = B(1-\frac{1}{2})(1-\frac{3}{2}) = \frac{1}{2} \cdot \frac{1}{2} \cdot B = \frac{1}{4}$ $B=-1$

$t=1/3$ $\frac{5}{9} - \frac{4}{3} + 1 = \frac{2}{9} = C(1-\frac{1}{3})(1-\frac{2}{3}) = \frac{2}{3} \cdot \frac{1}{3} \cdot C = \frac{2}{9}$ $C=1$

$G(t) = \frac{1}{1-t} - \frac{1}{1-2t} + \frac{1}{1-3t}$

$\sum_{n=0}^{\infty} a_n t^n = \sum_{n=0}^{\infty} t^n - \sum_{n=0}^{\infty} 2^n t^n + \sum_{n=0}^{\infty} 3^n t^n$

$a_n = 1 - 2^n + 3^n$

Ex. $a_n - 5a_{n-1} + 6a_{n-2} = 3^n \quad n \geq 2 \quad a_0 = 0 \quad a_1 = 2$

$$\sum_{n=2}^{\infty} a_n x^n - 5 \sum_{n=2}^{\infty} a_{n-1} x^n + 6 \sum_{n=2}^{\infty} a_{n-2} x^n = \sum_{n=2}^{\infty} 3^n x^n \quad \text{Sum from 2 to } \infty$$

$$(G(x) - a_0 - a_1 x) - 5x(G(x) - a_0) + 6x^2 G(x) = \sum_{n=2}^{\infty} 3^n x^n$$

$$G(1 - 5x + 6x^2) + a_0(5x - 1) - a_1 x = 3^2 x^2 \sum_{n=2}^{\infty} 3^{n-2} x^{n-2} \quad \begin{matrix} n-2=k \\ n=2 \quad k=0 \end{matrix}$$

$$G(1 - 5x + 6x^2) - 2x = 3^2 x^2 \sum_{k=0}^{\infty} 3^k x^k$$

$$G(1 - 3x)(1 - 2x) - 2x = \frac{9x^2}{1 - 3x}$$

$$G(1 - 3x)(1 - 2x) = \frac{9x^2}{1 - 3x} + 2x = \frac{3x^2 + 2x}{1 - 3x}$$

$$G = \frac{3x^2 + 2x}{(1 - 2x)(1 - 3x)^2} = \frac{A}{1 - 2x} + \frac{B}{1 - 3x} + \frac{C}{(1 - 3x)^2}$$

$$3x^2 + 2x = A(1 - 3x)^2 + B(1 - 3x)(1 - 2x) + C(1 - 2x)$$

$$x = 1/3 \Rightarrow 1 = 10A$$

$$x = 1/3$$

$$\frac{3 \cdot \frac{1}{9} + 2 \cdot \frac{1}{3}}{\frac{2}{3}} = \frac{C(1 - \frac{2}{3})}{\frac{2}{3}}$$

$$1 = \frac{C}{3} \quad C = 3$$

$$x = 1/2$$

$$3 \cdot \frac{1}{4} + 2 \cdot \frac{1}{2} = A(1 - \frac{3}{2})^2$$

$$7/4 = A/4 \quad A = 7$$

$$x = 0$$

$$0 = A + B + C \quad B = -(A + C) = -10$$

$$G = \frac{7}{1 - 2x} - \frac{10}{1 - 3x} + \frac{3}{(1 - 3x)^2}$$

$$\sum a_n x^n = 7 \sum 2^n x^n - 10 \sum 3^n x^n + 3 \sum (n+1) 3^n x^n$$

$$a_n = 7 \cdot 2^n - 10 \cdot 3^n + 3(n+1) 3^n$$

$$= 3^n (3n + 3 - 10) + 7 \cdot 2^n$$

$$= 3^n (3n - 7) + 7 \cdot 2^n$$

Ex.

$$a_n - 4a_{n-2} = n+1$$

$$a_0 = 0 \quad a_1 = 1 \quad n \geq 2$$

$$\sum_{n=2}^{\infty} a_n x^n - 4 \sum_{n=2}^{\infty} a_{n-2} x^n = \sum_{n=2}^{\infty} (n+1) x^n = \sum_{n=2}^{\infty} n x^n + \sum_{n=2}^{\infty} x^n$$

$$(G - a_0 - a_1 x) - 4x^2 G = \sum_{n=2}^{\infty} n x^n + \sum_{n=2}^{\infty} x^n$$

$$G - x - 4x^2 G = G(1 - 4x^2) - x = \sum_{n=2}^{\infty} n x^n + \sum_{n=2}^{\infty} x^n$$

$$G(1 - 4x^2) - x = \sum_{n=2}^{\infty} n x^n + x^2 \sum_{n=2}^{\infty} x^{n-2}$$

$$n-2 = k \quad n=2 \quad k=0$$

$$= \sum_{n=2}^{\infty} n x^n + x^2 \sum_{k=0}^{\infty} x^k$$

$$= \sum_{n=2}^{\infty} n x^n + \frac{x^2}{1-x}$$

$$G(1 - 4x^2) = \sum_{n=2}^{\infty} n x^n + \frac{x^2}{1-x} + x$$

$$G(1 - 4x^2) = \frac{2x^2 - x^3}{(1-x)^2} + x$$

$$= \frac{2x^2 - x^3}{(1-x)^2}$$

$$G = \frac{x + x^2 - x^3}{(1-x)^2(1-4x^2)} = \frac{x + x^2 - x^3}{(1-x)^2(1-2x)(1+2x)}$$

$$= \frac{A}{1-x} + \frac{B}{(1-x)^2} + \frac{C}{1-2x} + \frac{D}{1+2x}$$

$$= \frac{A}{1-x} + \frac{B}{(1-x)^2} + \frac{C}{1-2x} + \frac{D}{1+2x}$$

$$= \frac{-8}{9(1-x)} - \frac{1}{3(1-x)^2} - \frac{1}{36(1+2x)} + \frac{5}{4(1-2x)}$$

$$a_n = -\frac{8}{9} - \frac{1}{3}(n+1) - \frac{1}{36} 2^n + \frac{5}{4} (-2)^n$$

$$A(x) = \sum_{n=2}^{\infty} n x^n$$

$$A(x) = 2x^2 + 3x^3 + 4x^4 + \dots$$

$$xA = 2x^3 + 3x^4 + 4x^5 + \dots$$

$$A - xA = 2x^2 + x^3 + x^4 + \dots$$

$$= 2x^2 + x^3(1 + x + \dots)$$

$$= 2x^2 + \frac{x^3}{1-x}$$

$$= 2x^2 + \frac{x^3}{1-x}$$

$$= \frac{2x^2 - x^3}{1-x}$$

$$A = \frac{2x^2 - x^3}{(1-x)^2}$$

$$x + x^2 - x^3 = A(1-x)(1+2x) + B(1-4x) + C(1-x)^2$$

$$x + x^2 - x^3 = B(1-4x) \quad B = -1/3$$

$$x + x^2 - x^3 = A(1-x)(1-2x)(1+2x) + B(1-2x)(1+2x) + C(1-x)^2(1+2x) + D(1-x)^2(1-2x)$$

$$x=1 \quad 1 = B(1-2)(1+2) \quad B = -1/3$$

$$x=1/2 \quad \frac{1}{2} + \frac{1}{4} - \frac{1}{8} = C(1-1/2)^2(1+1)$$

$$\frac{4+2-1}{8} = C \cdot \frac{1}{4} \cdot 2$$

$$C = \frac{5}{4} \cdot \frac{4}{2} = 5/4$$

Q $a_n - 4a_{n-1} + 3a_{n-2} = 2^n$ $a_1 = 1, a_2 = 11$

Homo. $a_n - 4a_{n-1} + 3a_{n-2} = 0$

CE $r^2 - 4r + 3 = 0 = (r-1)(r-3) = 0$
 $r = 1, 3$

$a_n^h = b_1 \cdot 1^r + b_2 \cdot 3^r = b_1 + b_2 \cdot 3^r$

GenSol 2^n Trial Sol = $A \cdot 2^n$

Substitution

$A \cdot 2^n - 4A \cdot 2^{n-1} + 3A \cdot 2^{n-2} = 2^n$

$\frac{A}{2} - \frac{4A}{1} + \frac{3A}{4} = 1$

$\frac{A - 8A + 3A}{4} = 1$

$A = -4$

$a_n^h = -4 \cdot 2^n$

GenSol. $= a_n = b_1 + b_2 \cdot 3^n - 4 \cdot 2^n$

$a_1 = 1 = b_1 + 3b_2 - 8$ $b_1 + 3b_2 = 9$

$a_2 = 11 = b_1 + 9b_2 - 16$ $b_1 + 9b_2 = 27$

$6b_2 = 18$

$b_2 = 3$

$b_1 = 0$

GenSol

$a_n = 3 \cdot 3^n - 4 \cdot 2^n = 3^{n+1} - 2^{n+2}$

$$2a_n - 7a_{n-1} + 3a_{n-2} = 2^n$$

$$\text{CE} \quad 2n^2 - 7n + 3 = 0$$

$$2n^2 - n - 6n + 3$$

$$n(2n-1) - 3(2n-1) = 0$$

$$(n-3)(2n-1) = 0$$

$$n = 3, 1/2$$

$$a_n^h = b_1 \cdot 3^n + \frac{b_2}{2^n}$$

$$f(n) = 2^n$$

$$\text{Trial sol} = A \cdot 2^n$$

Subst

$$2A2^n - 7A2^{n-1} + 3A2^{n-2} = 2^n$$

$$\frac{2A2^n - 7A2^n}{2} + \frac{3A2^n}{4} = 2^n$$

$$8A - 14A + 3A = 4$$

$$-3A = 4$$

$$A = -4/3$$

$$a_n^p = -\frac{4}{3} 2^n$$

$$a_n = b_1 3^n + \frac{b_2}{2^n} - \frac{4}{3} 2^n$$